We have proved exactly what we set out to prove: the momentum is equal to the mass times the velocity, or, more exactly, the average momentum equals the mass times the velocity.

To get a better idea of what this means, let's suppose the wave function has the form of a packet, or fairly narrow lump. The expectation value of x will be approximately at the center of the lump. What Eq. 9.11 tells us is that the center of the wave packet travels according to the classical rule $p = mv$.

9.5 QUANTIZATION

Before moving on to the subject of forces in quantum mechanics, I want to pause and discuss what we have done. We started with a well-known and well-trusted classical system—the free particle—and *quantized* it. We can codify this procedure as follows:

1. Start with a classical system. This means a set of coordinates x and momenta p. In our example, there was only one coordinate and one momentum, but the procedure is easy to generalize. The coordinates and momenta come in pairs, x_i and p_i. The classical system also has a Hamiltonian, which is a function of the x's and p's.

2. Replace the classical phase space with a linear vector space. In the position representation, the space of states is represented by a wave function $\psi(x)$ that depends on the coordinates—in general, all of them.

3. Replace the x's and p's with operators $\mathbf{X}_i$ and $\mathbf{P}_i$. Each $\mathbf{X}_i$ acts on the wave function to multiply it by x_i. Each $\mathbf{P}_i$ acts according to the rule

$$\mathbf{P}_i \rightarrow -i\hbar \frac{\partial}{\partial x_i}.$$